E-commerce Checkout Drop-off Analysis

Business Problem Document

Project Context

An e-commerce business generates millions of product-view, cart and purchase events, but a substantial share of shopping sessions do not progress from cart activity to purchase. Management needs a reliable way to quantify checkout drop-off, understand behavioral patterns associated with abandonment, and identify product categories where intervention is most commercially important.

Business Problem

The organization lacks a consolidated view of the customer journey from product exploration to cart and purchase. Without session-level funnel metrics, abandonment drivers and category prioritization, teams may optimize the wrong areas or focus only on high abandonment percentages without considering business volume.

Primary Objective

Analyze e-commerce event data using PostgreSQL and Power BI to measure cart abandonment, identify behavioral and category-level patterns associated with drop-off, and translate the findings into testable business actions.

Business Questions

• How many sessions view products, add products to cart and complete a purchase?

• What is the cart-to-purchase conversion rate and cart abandonment rate?

• How does abandonment vary by cart size, product exploration, repeated product views and time of day?

• How do abandoned sessions differ from converted sessions in duration and product-view behavior?

• Which product categories combine high abandonment with meaningful cart volume and customer friction?

• Which opportunities should the business prioritize first, and how should success be measured?

Scope and Analytical Unit

The raw dataset contains event-level activity. The analysis converts these events into session-level and category-level reporting structures. A session is the core behavioral unit. Cart abandonment is evaluated only among sessions with cart activity, preventing view-only sessions from being incorrectly classified as abandoned carts.

Core KPIs

KPI	Definition / Business Meaning
Total Sessions	Distinct customer sessions in the analysis.
Cart Sessions	Sessions containing at least one add-to-cart event.

KPI	Definition / Business Meaning
Converted Cart Sessions	Cart sessions that also contain a purchase.
Abandoned Cart Sessions	Cart sessions with no purchase.
Cart Abandonment Rate	Abandoned cart sessions / all qualifying cart sessions.
Cart-to-Purchase Rate	Converted cart sessions / all qualifying cart sessions.
Avg. Session Minutes	Average session duration used to compare behavior.
Avg. Product Views	Average product-view events per session.
Category Abandonment Rate	Abandoned cart sessions within a category / category cart sessions.
Repeat View Rate	Behavioral friction indicator based on repeated product exploration.

Stakeholders

• E-commerce / Product Manager - checkout and product-experience decisions.

• Growth / CRM Team - cart recovery and high-intent customer targeting.

• Category Managers - product-category prioritization and merchandising actions.

• UX / CRO Team - checkout simplification and decision-support experiments.

• Analytics / BI Team - KPI monitoring, segmentation and experiment measurement.

Expected Deliverables

• PostgreSQL analysis workflow and reusable reporting tables.

• Three-page Power BI dashboard: Executive Overview, Abandonment Drivers, Category Opportunities.

• Business recommendations framed as hypotheses and experiments.

• Portfolio-ready report, presentation and GitHub documentation.

Success Criteria

Stakeholders should be able to quantify the abandonment problem, understand where customer friction is concentrated, identify priority categories, and define measurable experiments rather than relying on assumptions.

Important Analytical Limitation

The dataset is observational event data. The analysis identifies associations, not causal effects. The dataset also does not expose every real-world checkout factor such as payment failure, shipping fees, device type, traffic source or explicit reason-for-exit fields.